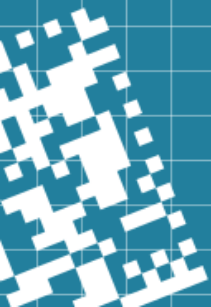
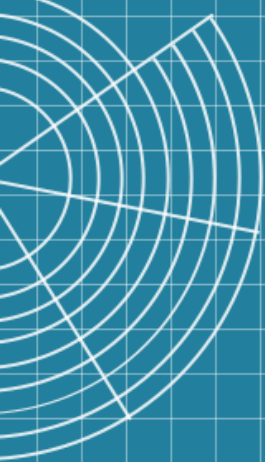
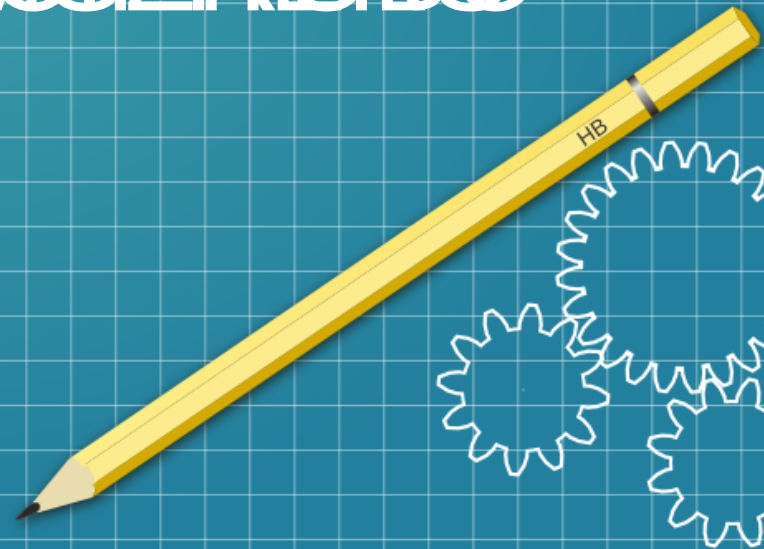
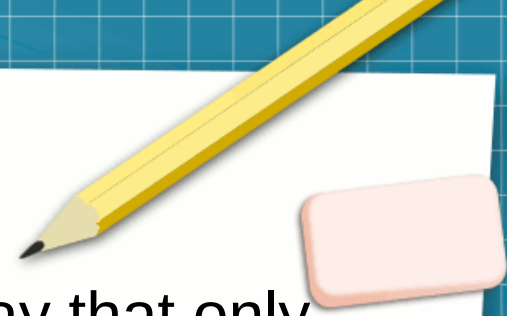


EXNORZDSNOIAXNDMSZREBS

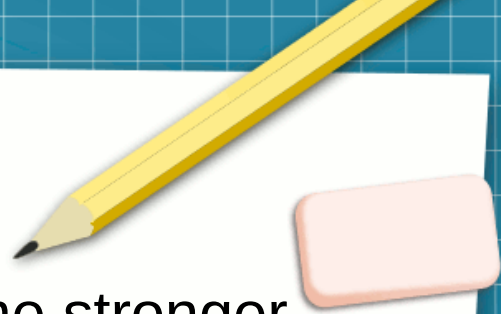


What is Encryption



- Its the process of transforming information in a way that only authorised parties can read.
- The original code/words/information known as `plaintext`
- The encrypted version is known as `cyphertext`
- To get `cyphertext` the `plaintext` is Encrypted through an encryption algorithm.

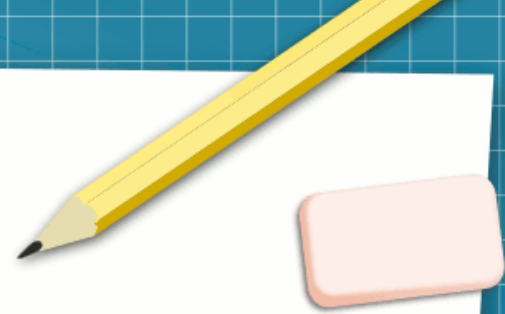
Encryption



- There are many types of encryption algorithms, some stronger than others.
- The encryption algorithm is usually based upon a `key` or a piece of information only known by the person/computer encrypting the data.
- The key is usually `random` number/characters
- The more `random` and longer the `key` the stronger the encryption.
- The process is reversible

Decryption

- The opposite to Encryption
- Taking `cyphertext` and transforming it to `plaintext`
- You need the `key` to decrypt the information



Why do we need Encryption?

- Privacy (Messaging Apps – WhatsApp, Signal)
- Web browsing (HTTPS)
- Online Banking
- Contactless Payments
- Mobile Phones
- Satellites
- Laptops/Desktops/Servers
- WiFi (PSK – Pre-Shared Key, this is your WiFi Password)



Ciphers



- Ciphers are the `Method` or `Algorithm` used to perform Encryption
- Examples:
 - AES (Advanced Encryption Standard)
 - DES (Data Encryption Standard)
 - RSA (Rivest, Shamir, Adleman Algorithm)
 - SHA (Secure Hash Algorithm)

Early Ciphers



- Ciphers have been around a very long time, even before computers
- Ancient Greeks are known to have used it.
 - Scytale which used Transposition*
- Ancient Roman used it, Popularly by Julius Caesar
 - Caesar Cipher – which was a Substitution cipher
- Germany used it in World War II
 - The Enigma machine

Transposition Cipher



Key	C	I	P	H	E	R
Sequence	1	4	5	3	2	6
Plaintext	H	E	L	L	O	M
	Y	N	A	M	E	I
	S	O	L	I	*	*

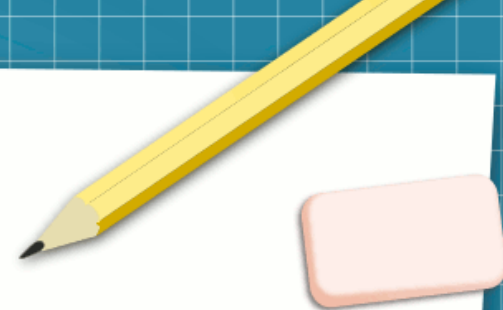
Ciphertext by Column in Sequence: #1 HYS #2 OE #3 LMI #4 ENO #5 LAL #6 MI

Ciphertext : HYS OE LMI ENO LAL MI

Plaintext : HELLO MY NAME IS OLI

Can you work out the plaintext?

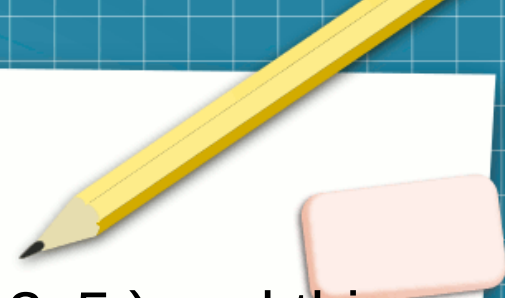
Decrypt



Key	S	E	C	R	E	T	S
Sequence	1	6	3	2	4	5	7
Plaintext	T	O	D	A	Y	W	E
	L	E	A	R	N	C	I
	P	H	E	R	S	*	*

Ciphertext: TLP ARR DAE YNS WC OEH EI

Route Cipher



- The plaintext is written in a grid of a certain size (3x5) and this would be part of the `key`
- The Cipher text would then be read in a pattern (row by row, spiral etc)

T	S	M	E	E
H	I	Y	C	T
I	S	S	R	!

Ciphertext: TSMEE HIYCT ISSR!

Plaintext: THIS IS MY SECRET!

Decrypt

- Ciphertext: !!NU FSIG NITU PMOC
- Key: 4x4 – right to left – bottom to top

!	!	N	U
F	S	I	G
N	I	T	U
P	M	O	C

Plaintext: COMPUTING IS FUN !!



Caesar Cipher

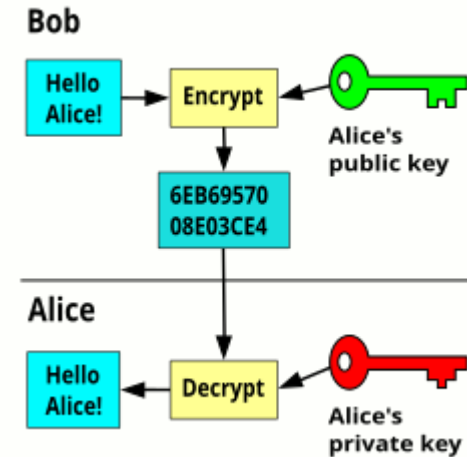
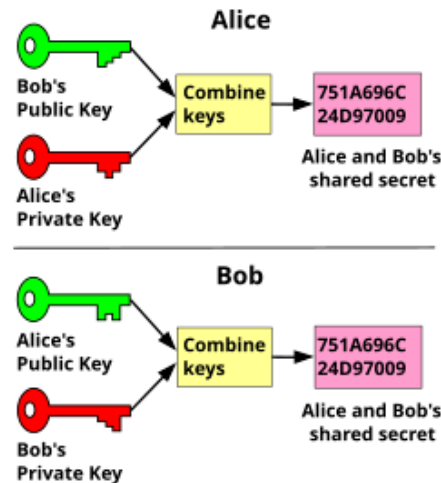


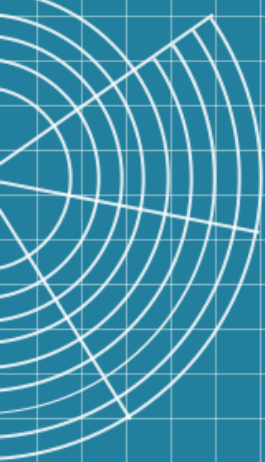
- A Caesar Cipher is a substitution Cipher and is one of the simplest.
- It replaces characters by shifting the characters a fixed number, these are known as shifts, The below Cipher is shift right 11

Shift	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
Plain	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P

Public Key Cryptography

- This type of encryption is what secures the internet today.
- Relies on a large random number and a key generator algorithm to produce a `Public Key` and a `Private Key`





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